Singapore rainfall: temporal-spatial patterns and prediction

Tasks 1 and 2: Spark exploration and probability prediction, 2017–2024

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Study objectives. Using Singapore rainfall observations from 2017–2024, we analyse temporal and spatial patterns and develop models to predict the probability that station rainfall accumulated over the next 30 minutes exceeds a given threshold.

Data and methods. The eight NEA / data.gov.sg annual datasets contain 48,538,713 source records and 48,538,627 unique valid observations from 91 station IDs. Spark performs full-data cleaning and aggregation. Annual comparisons use 21 stable stations to reduce changes caused by the monitoring network.

Main findings. Of all valid five-minute observations, 97.08% are dry. Eligible diurnal samples show the highest rain frequency at 16:00–17:00 Singapore time. Available-year monthly averages have the highest frequency in November and the lowest in July, with varying eligible years and stations. In the fixed panel, duration-normalized rainfall is lowest in 2019 and highest in 2021. Nearby stations generally have more strongly correlated daily rainfall; eight years alone do not establish long-term climate change.

Prediction findings. Models use all 34,542,285 eligible 2017–2022 records for training, 2023 for calibration and selection, and 2024 for testing. The original comparison selected three XGBoost models. Subsequent 5 mm tuning increased test PR-AUC from 0.21839 to 0.22640 and reduced Brier from 0.0072009 to 0.0071609. The prediction interface uses this tuned 5 mm model and retains the original 0 and 25 mm models. Additional spatial interactions did not improve 2023 validation performance and were not adopted.

Year	Rain / observed 24h (mm)	Wet frequency (%)	Wet-interval intensity (mm/5min)	Mean coverage (%)
2017	6.85	2.85	0.833	94.6
2018	7.10	2.90	0.840	82.9
2019	4.81	1.97	0.841	94.7
2020	7.40	2.92	0.877	95.3
2021	7.84	3.04	0.895	97.8
2022	7.71	3.08	0.868	99.0
2023	7.52	3.16	0.824	95.2
2024	7.11	3.01	0.819	95.6

Each year in this table uses the same 21 stations, with equal station weights. Duration-normalized rainfall is neither an official annual total nor an actual daily total. Eligible monthly and hourly samples can vary.

1. Data sources, fields and cleaning

Sources and fields. Records contain observation and update timestamps, station ID/name/device ID, coordinates, reading update timestamp, rainfall amount, type, unit and date fields. There are no temperature, humidity or wind observations. API previews identify fields and annual IDs; weather findings come from the complete annual files.

Data checks. After downloading complete annual CSV files, we reconcile Spark input row counts with annual API metadata and check duplicate observations, timestamp alignment and rainfall values.

Cleaning. Spark parses timestamps in Asia/Singapore. It accepts finite, nonnegative mm readings with the expected five-minute type and valid station ID, timestamp and CSV record. Timestamps are aligned only if within two seconds of the five-minute grid; original epochs are retained. Station-time duplicates use the latest reading update, then deterministic tie-breakers. Combined annual data are checked again for cross-file duplicates.

Year	Source rows	Duplicates removed	Off-grid over 2 s	Negative readings	Stations
2017	5,256,106	86	0	0	62
2018	4,594,246	0	0	0	56
2019	5,026,862	0	0	0	54
2020	6,334,473	0	0	0	80
2021	7,044,646	0	0	0	76
2022	7,133,230	0	0	0	72
2023	6,742,890	0	0	0	74
2024	6,406,260	0	0	0	70

Cleaning results. All 86 duplicate records occur in 2017. The eight years contain no negative rainfall, timestamps more than two seconds from the five-minute grid, or cross-file duplicates. Missing intervals are not counted as zero rainfall. Additional screening of readings above 50 mm/5min identifies none.

2. Metrics and comparable samples

For a station and time period, let N be the number of valid observations, $r_i \geq 0$ the rainfall amount, $W = \sum_i \mathbf{1}(r_i > 0)$ the number of wet intervals, and E the expected calendar slots. We use:

$$\begin{aligned} \text{Coverage} &= N/E, \qquad E = 288 \times \text{calendar days}, \\ \text{Wet frequency (\%)} &= 100W/N, \\ \text{Wet-interval intensity (mm/5min)} &= \sum_i r_i/W \quad (W > 0), \\ \text{Rain / observed 24h (mm)} &= 288 \sum_i r_i/N. \end{aligned}$$

Calendar denominators handle leap years. Low coverage can reflect a station operating for only part of a year. Intensity averages wet intervals only. For one station and period, normalized rainfall equals 288 × frequency/100 × wet-interval intensity. This product generally does not hold after separate station averaging. Normalized rainfall is not an official daily or annual total and does not reconstruct missed storms.

Fixed annual panel. Of 91 station IDs across eight years, 21 satisfy all criteria: observations in every year, annual coverage of at least 80% in each year, and latitude/longitude spans no larger than 0.005°. The other 70 are excluded only from fixed-panel annual comparisons; their valid observations were not deleted during cleaning. The 21 stations receive equal weights. The next page lists the station scope for each analysis.

Monthly and diurnal samples. Figures 2 and 4 further screen sufficiently complete months or hourly strata from the 21-station pool. Monthly summaries average stations equally, then available years equally. Each eligible year-month has 17–21 stations, and each month has 6–8 available years. Requiring the same station-month to qualify in all eight years leaves only May, June, July, September, October and December, used as an additional comparison.

Spatial samples. Figure 1 uses 2024, which has 70 stations. Of these, 56 have annual coverage of at least 90% and enter map colouring; the other 14 are shown in grey. Figure 5 uses daily totals from the fixed 21 stations, requiring at least 95% daily coverage and 365 common valid days per pair. The 80%, 90% and 95% completeness thresholds are project choices.

3. Computing environment and station scope

Computing environment. Task 1 runs on an Apple Silicon Mac with 10 logical CPUs, 16 GiB RAM, macOS 26.5.2 ARM64, Python 3.12.14, Java Temurin 17.0.20.1+1 and Spark 3.5.8. Task 2 runs on gpushare with Python 3.11.10, Azul Zulu Java 17.0.20.1, Spark 3.5.8 and XGBoost 3.2.0. The gpushare container quota is 42 CPUs and about 667.68 GiB RAM; full-feature model training uses 20 CPUs and a 6 GiB default driver heap.

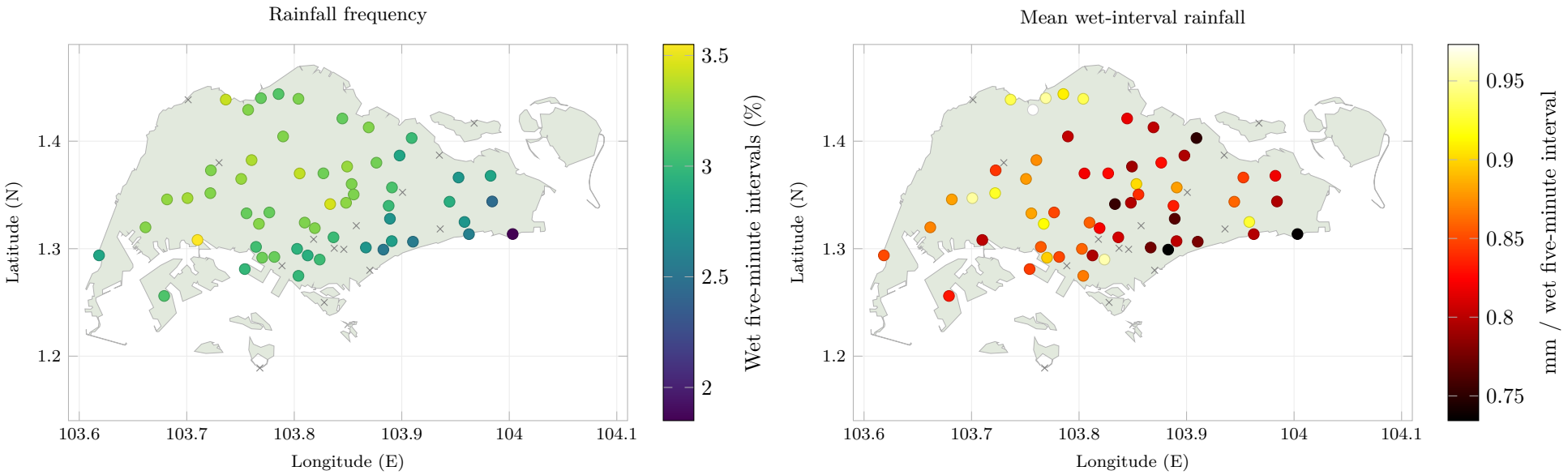
Spark configuration and responsibilities. Task 1 and the initial feature-group comparisons use single-machine `local[2]`. Full-feature model comparison and evaluation use `local[20]`, 80 shuffle partitions and default parallelism 20. XGBoost uses two workers with ten CPUs per task. Experiments run sequentially at low priority without a GPU. Spark DataFrame/SQL and windows process full-data cleaning, aggregation, labels and features; Spark ML pipelines train and evaluate models. Ordinary phases use a 6 GiB Java heap and a 12 GiB process-tree stop limit. XGBoost fitting and evaluation have a 40 GiB process-tree limit.

Analysis samples. Station scope and completeness criteria are chosen according to the comparison, as summarized below.

Analysis	Station scope	Screening rule
Figure 1: 2024 station map	56 of 70 stations that year	Annual coverage $\geq 90\%$; 14 shown in grey
Figure 2: monthly patterns	Eligible subsets of the fixed 21	17–21 stations per eligible year-month
Figure 3: annual changes	The same 21 stations	Other 70 excluded from this comparison only
Figure 4: diurnal patterns	Eligible strata of the fixed 21	Year-month-hour coverage $\geq 80\%$
Figure 5: distance and correlation	Fixed 21 stations; 210 pairs	Daily coverage $\geq 95\%$; ≥ 365 shared days
Figure 6 left: wet amounts	All 91 stations across eight years	Wet five-minute observations only
Figure 6 middle/right: exceedance and persistence	Valid 2024 observations from all 70 stations	Six complete future slots or two adjacent slots

Cleaning checks. Source row counts match API counts; 86 duplicate records were removed, and missing slots are not zero-filled. The overall dry fraction uses all 91 stations, rather than only the 21-station panel.

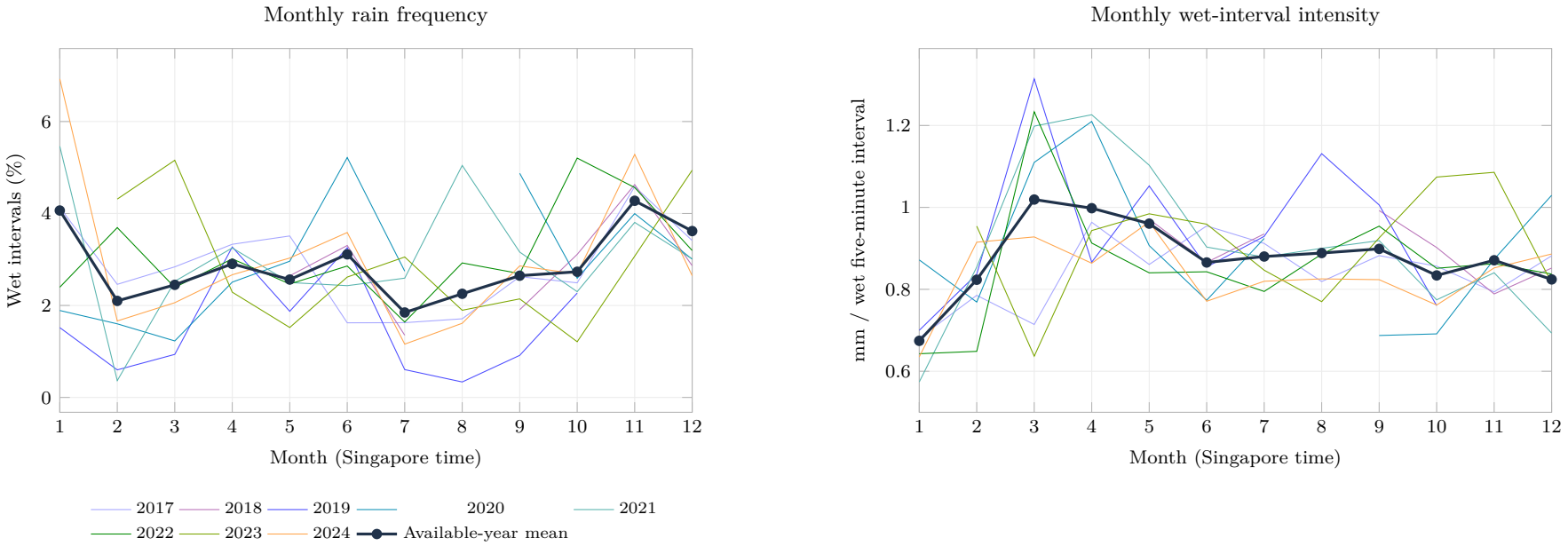
Figure 1: Station rainfall maps in 2024



In 2024, 56 of 70 stations have annual coverage of at least 90%. Coloured points show individual gauge statistics; grey crosses indicate lower coverage. S33 (Jurong Pier Road) has the highest rain frequency, 3.55%. No spatial interpolation is performed. The geoBoundaries land outline is derived from URA 2016 data under ODbL 1.0 and serves as a geographic reference.

Task 2 feature implications (stations). Candidate features include station ID and coordinates to represent station differences. Task 2 should screen windows for historical-feature and future-label completeness, rather than automatically restrict training to the 21-station annual panel.

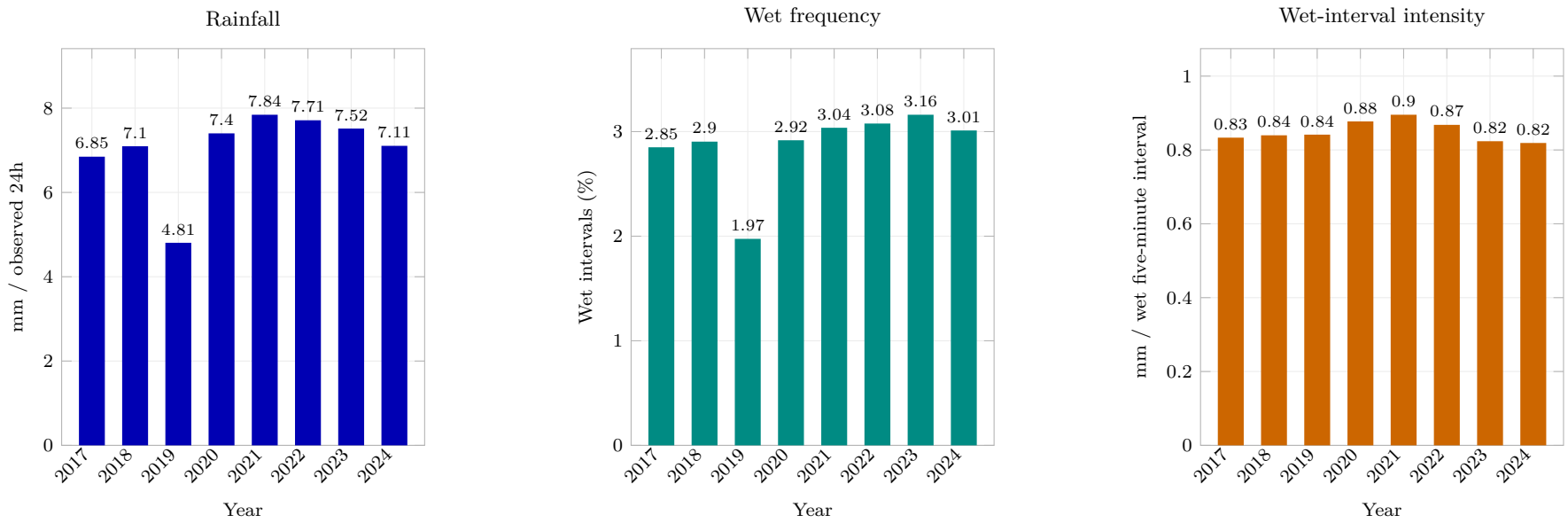
Figure 2: Monthly patterns within a year



Eligible station-months from the fixed panel require at least 80% coverage, with equal averaging over stations followed by available years. Mean monthly rain frequency is highest in November (4.28%) and lowest in July (1.85%). Mean wet-interval amounts are about 1.02 mm in March and 0.67 mm in January. Frequency and wet-interval amount show different seasonal patterns. Each month includes 6–8 years, and each eligible year-month includes 17–21 stations.

Task 2 features (month). Monthly frequency and wet-interval intensity vary differently. We encode month using sine and cosine to represent the annual cycle and combine them with hour encoding to capture seasonal variation in diurnal rainfall.

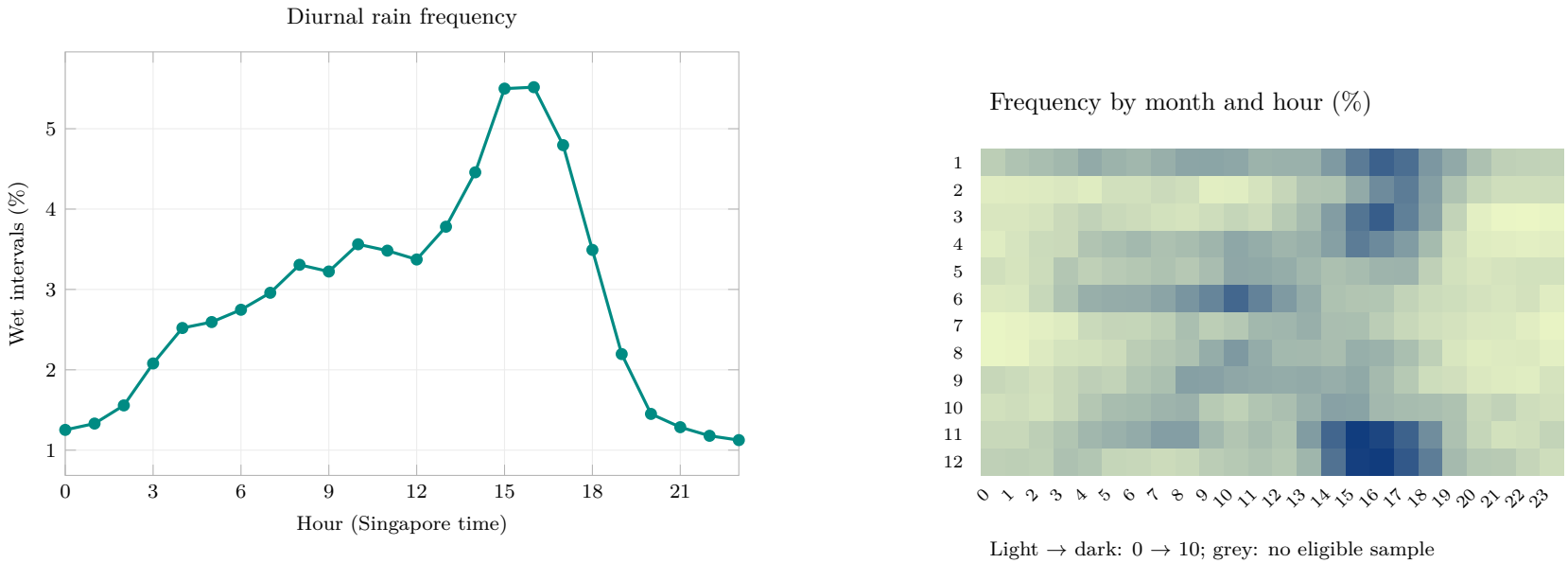
Figure 3: Annual changes in the fixed panel



Each year compares the same 21 equally weighted stations, each with annual coverage of at least 80%. Mean rainfall per observed 24 hours ranges from 4.81 mm in 2019 to 7.84 mm in 2021. Rain frequency in 2019 is 1.97%, below the other years, while its wet-interval amount of 0.841 mm is close to adjacent years. Lower rainfall frequency is thus the main component associated with its lower normalized rainfall.

Task 2 features (year). Training, validation and test are split chronologically. Models use cyclic month and hour features without year identifiers. NEA relates the dry conditions in 2019 to a strong positive Indian Ocean Dipole, providing external context for annual variation.

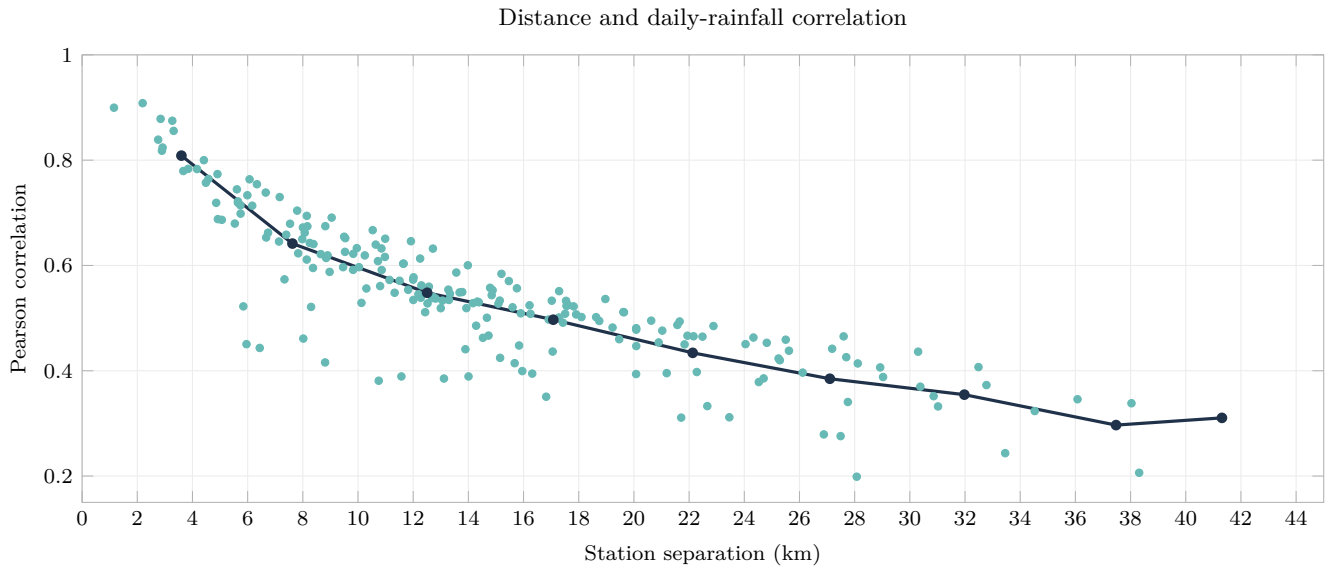
Figure 4: Diurnal variation in rainfall frequency



Rain frequency peaks at 16:00–17:00, with peak timing and strength varying by month. The afternoon peak is more pronounced in November–December, while June has elevated frequency from morning toward midday. Statistics equally weight eligible station-year-month-hour strata with at least 80% coverage. These findings motivate hour, month and their interactions as predictors.

Task 2 feature implications (hour and month). Encode Singapore local hour cyclically and consider month–hour interactions: the strength and timing of the afternoon peak vary by month. Logistic regression and linear SVM need explicit interaction features; trees can learn such combinations. Their predictive contribution must still be tested by ablation.

Figure 5: Distance and daily-rainfall correlation

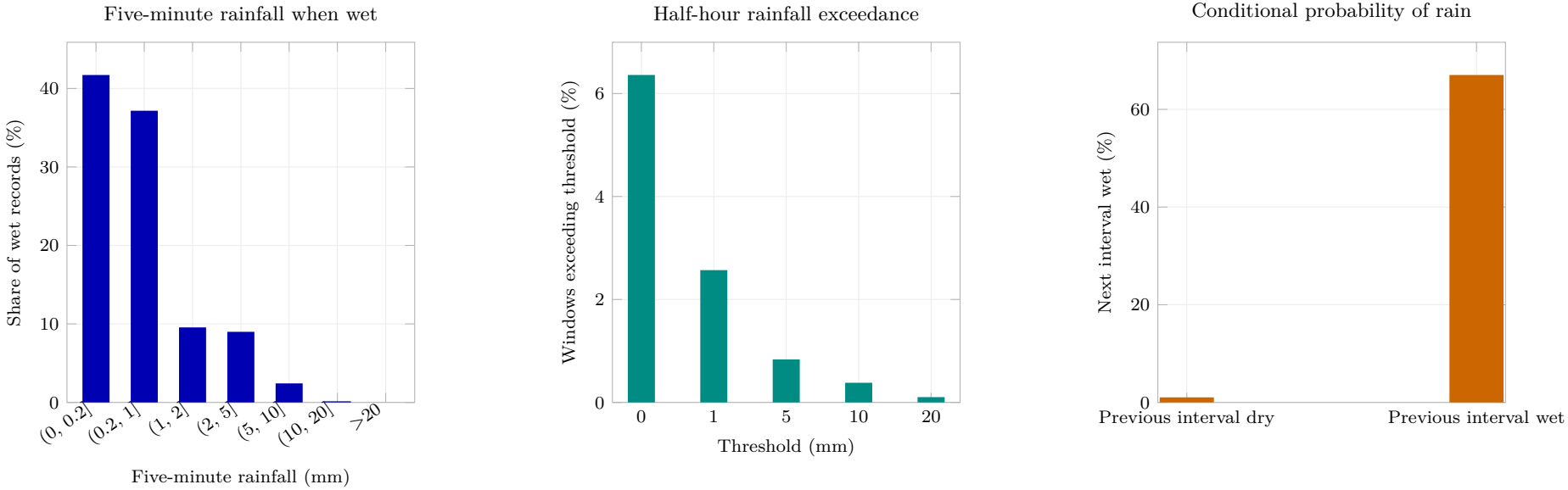


The 21 stations form 210 pairs. The mean Pearson correlation of daily rainfall is 0.809 for 17 pairs separated by at most 5 km, compared with 0.336 for 12 pairs more than 30 km apart. Mean correlations in 5 km distance groups generally decrease with separation, indicating spatial dependence.

Stations are not independent. Nearby gauges may experience the same rainfall event, and different pairs can share a station. Thus, 91 gauges are not 91 independent weather samples, and the 210 correlations are not 210 independent samples. Shared-day screening follows the criteria given earlier.

Task 2 features (neighbours). Stronger daily-rainfall correlation among nearby stations motivates mean rainfall, maximum rainfall and wet fractions from the three closest stations. Task 2 compares models before and after adding neighbour features to evaluate their contribution to half-hour prediction.

Figure 6: Rainfall distribution, half-hour exceedance and persistence



Rainfall distribution in wet intervals. Across all stations and eight years, 41.72% of wet five-minute observations fall in $(0, 0.2]$ mm and 37.16% in $(0.2, 1]$ mm. Together, 78.88% have at most 1 mm, indicating small accumulations in most wet intervals. Dry intervals account for 97.08% of all valid five-minute observations.

Exceedance probability of half-hour rainfall. There are 6,374,476 complete future six-interval windows in 2024. Subsequent 30-minute totals strictly exceed 0 mm in 6.36% of windows and 5 mm in 0.83%; exceedance becomes less frequent as the rainfall threshold rises. Totals are compared at nine decimal places to avoid floating-point errors at threshold boundaries. The horizontal axis uses threshold categories.

Rainfall persistence in consecutive intervals. For consecutive five-minute observations at the same station in 2024, the probability of rain is 1.04% following a dry interval and 67.03% following a wet interval. This substantial short-term persistence supports recent rainfall and wet-spell duration as predictors.

Task 2 feature implications (persistence and rare events). Implemented features include the latest reading, 15/30/60-minute totals, recent rainfall changes and wet-spell duration. Only records available at prediction time are used; missing readings are not dry readings. The following sections report positive counts, PR-AUC, Brier score and calibration curves separately by threshold.

4. Task 2: target and chronological splits

Prediction target. For station s, time t and rainfall threshold q, estimate the probability that rainfall in the next 30 minutes strictly exceeds q. Use $q \in \{0, 5, 25\}$ mm and one classifier per target, with one feature row and three label columns per station/time.

Threshold choices. We preset rainfall thresholds of 0, 5 and 25 mm to examine any rainfall, exceedance of a lower positive accumulation threshold, and a larger accumulation event over the next 30 minutes. These targets allow comparison of model ranking and probability quality as events become rarer. The thresholds are experimental choices, not official rainfall categories. A classifier is trained and selected separately for each threshold; the current interface supports only these three values rather than arbitrary continuous thresholds.

$$R^{30}_{s,t} = \sum_{k=1}^{6} r_{s,t+5k\,\min}, \qquad y^{(q)}_{s,t} = \mathbf{1}(R^{30}_{s,t} > q), \qquad \hat{p}_q = P(y^{(q)} = 1 \mid x_{s,t}).$$

Label construction. Treat timestamps as five-minute interval ends. Sum six valid intervals in $(t, t+30\text{min}]$ and round to nine decimal places before comparison. Exclude the current interval and reject windows missing any future interval. Equality with the threshold is negative.

Purpose	Rows	> 0 positives	> 5 positives	> 25 positives
2017–2022: all training records	34,542,285	2,074,430	279,801	19,787
2023 H1: calibration	3,325,499	251,794	30,699	1,532
2023 H2: selection	3,393,373	197,358	28,000	2,487
2024: test	6,374,476	405,430	53,193	3,517

Training and validation. All base classifiers and preprocessing use all 34,542,285 eligible 2017–2022 training records. Fit calibration on 2023 H1, select models and decision cutoffs on H2, and test on 2024. Label windows cannot cross these boundaries. Calibration, selection and test preserve natural class proportions.

5. From Task 1 to feature engineering

Task 1 finding	Construction	Inputs
Afternoon peak varies by month	Cyclic hour/month and interactions	8 calendar features; no year ID
Station differences and spatial correlation	Station ID, coordinates, three closest neighbours	Neighbour rain means, maxima and wet fractions
Short-term persistence	Latest reading and 5/10/15/30/60-minute lags	15/30/60-minute totals, changes and wet-spell length

History and neighbours. Use only observations available by prediction time: both observation and retained update timestamps must be no later than t. Search the past 90 minutes; mark the latest reading unavailable after 30 minutes. Use coordinates known by t to identify three closest eligible neighbours and aggregate rainfall, available counts, ages and distances. Spark windows and joins process all records.

Cyclic time encoding. Let h be local hour including fractional minutes and m the month. Use the cyclic encodings below and their four pairwise products, giving eight features. Adjacent 23:00/00:00 and December/January stay close in feature space.

$$h_{\sin} = \sin(2\pi h/24), \quad h_{\cos} = \cos(2\pi h/24), \qquad m_{\sin} = \sin[2\pi(m-1)/12], \quad m_{\cos} = \cos[2\pi(m-1)/12].$$

Input processing. Impute missing values using training-period means; select missingness indicators and fit the station vocabulary on the complete training period only. Use one-hot station encoding for linear models and a categorical index for trees; exclude constant features. Logistic and SVM standardize features. Nine logistic models compare calendar/station, added own history, and added neighbour history.

Feature comparison settings. The nine ablation classifiers use L2 parameter 0.001, at most 80 iterations and tolerance 10^{-6}. 0 reach this iteration cap. Logistic and SVM in the main algorithm comparison allow 100 iterations.

Feature dimensions. The complete input has 45 numeric features and 27 missingness indicators. With station encoding, linear models have 162 dimensions and tree models 73.

L2 logistic features	Test PR-AUC: 0 mm	5 mm	25 mm
Calendar/station	0.11588	0.01671	0.00139
Add own history	0.53158	0.15868	0.01482
Add neighbours	0.56134	0.17982	0.01713

Neighbour contribution. Adding neighbour history changes test PR-AUC by +0.02975, +0.02114, +0.00231 relative to own-history models, consistent with Task 1 spatial correlations.

6. Models, imbalance and calibration

Model	Parameters
L2 logistic (MLlib)	$\lambda = 0.001$, standardization, 100 iterations, tolerance 10^{-6}
Linear SVM (MLlib)	$\lambda = 0.001$, standardization, 100 iterations, tolerance 10^{-6}
XGBoost (Spark interface)	80 trees, depth 4, learning rate 0.1, 64 bins, minimum child weight 20; $L_2 = 1$

Training policies. Compare natural distribution, balanced weighting and constrained SMOTE. Let N be the number of real training records and P the target-specific positive count. Weights are $w_+ = N/(2P)$ and $w_- = N/[2(N - P)]$. SMOTE targets 20% positives and caps their total at ten times the original count. Every policy retains all real training observations; synthetic rows enter training only.

SMOTE construction. All training positives are available as donors, processed sequentially by station to bound memory. Search up to five neighbours with identical station, missingness, discrete profile and rain-zero pattern. Interpolate 16 continuous rain features with one coefficient, preserving calendar and discrete information; copy isolated donors.

Probability calibration. Fit Platt calibration $\hat{p} = 1/[1 + \exp(-(az + b))]$ on 2023 H1, where z is the base score and $a \geq 0$. This converts SVM scores to probabilities and adjusts weighting/resampling probability scales. A group predicted near 10% should ideally contain about 10% actual positives.

7. Evaluation and validation selection

Evaluation metrics. ROC-AUC measures ranking, with random-ranking reference 0.5. PR-AUC combines precision and recall and is useful for rare positives; random-ranking precision is approximately prevalence π. Spark integrates both areas exactly. Brier score is mean squared probability error; lower is better.

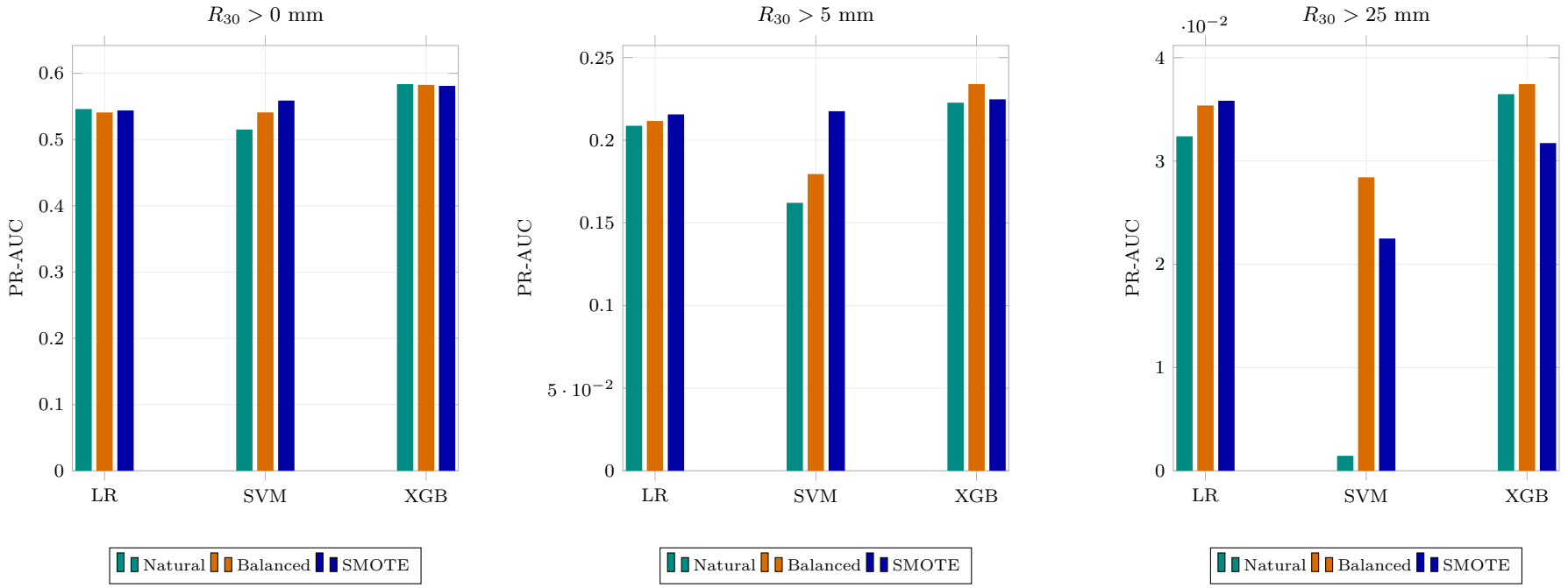
Model selection. Compare policies for each algorithm/threshold on 2023 H2, select primarily by PR-AUC and break ties with Brier. Choose decision cutoffs by validation F1 for test precision, recall, F1 and accuracy; these differ from rainfall thresholds defining labels. Test results do not select models.

Algorithm / policy	Selection PR-AUC: 0 mm	5 mm	25 mm
Logistic / Natural	0.54605	0.20874	0.03239
Logistic / Balanced	0.54088	0.21162	0.03537
Logistic / SMOTE	0.54389	0.21562	0.03583
SVM / Natural	0.51511	0.16211	0.00145
SVM / Balanced	0.54095	0.17952	0.02842
SVM / SMOTE	0.55882	0.21753	0.02249
XGBoost / Natural	0.58363	0.22270	0.03647
XGBoost / Balanced	0.58252	0.23392	0.03744
XGBoost / SMOTE	0.58096	0.22471	0.03173

Iteration status. Among the nine main logistic models, 0 reach the 100-iteration cap; 9 of the nine SVM models reach that cap.

Selected models. Validation selects 0 mm: Natural XGBoost; 5 mm: Balanced XGBoost; 25 mm: Balanced XGBoost. Evaluate all 9 algorithm/threshold policy winners on the full test set.

Figure 7: training policies on validation



Validation comparison. Relative to natural-distribution training, weighting improves PR-AUC in 7 of 9 algorithm/target combinations and SMOTE in 6. Evaluate imbalance treatment by algorithm and target. Every comparison uses the same complete training period and identical validation observations.

8. Test results in 2024

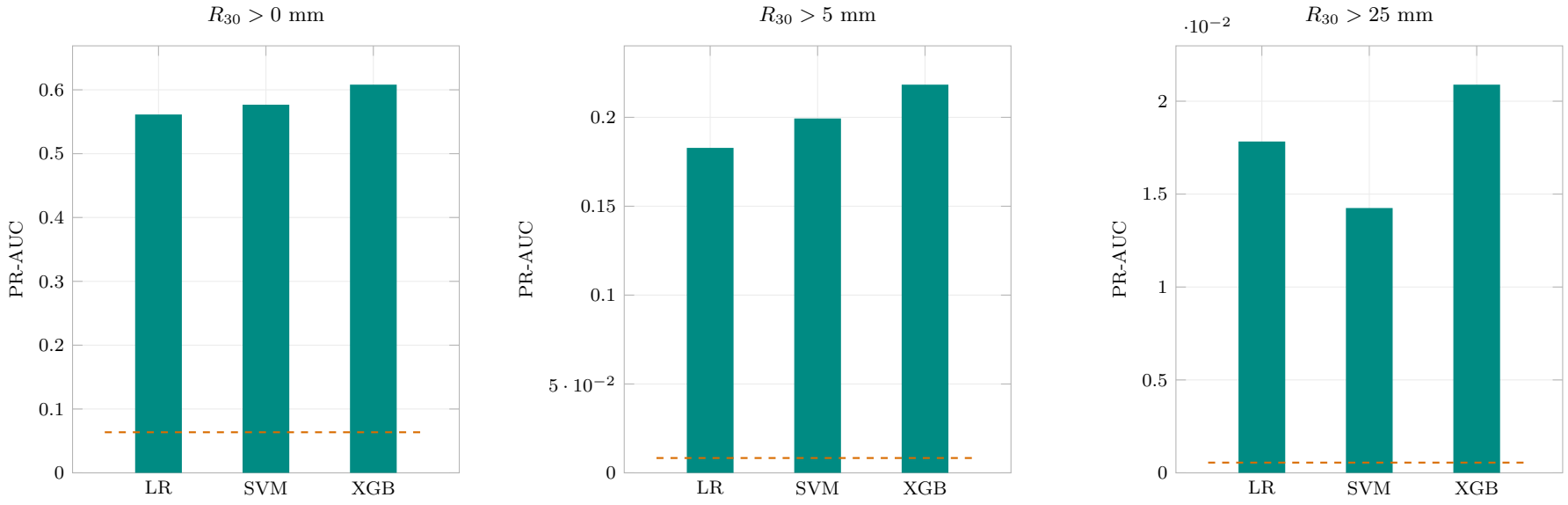
Threshold mm	Algorithm	Selected policy	ROC-AUC	PR-AUC	Brier
0	Logistic	Natural	0.85625	0.56134	0.0392597
0	SVM	SMOTE	0.85451	0.57665	0.0401863
0	XGBoost	Natural	0.87908	0.60809	0.0364909
5	Logistic	SMOTE	0.85293	0.18282	0.0075397
5	SVM	SMOTE	0.72667	0.19935	0.0077310
5	XGBoost	Balanced	0.87952	0.21839	0.0072009
25	Logistic	SMOTE	0.86018	0.01783	0.0005496
25	SVM	Balanced	0.85936	0.01425	0.0005487
25	XGBoost	Balanced	0.90435	0.02089	0.0005449

Test observations. Each target uses 6,374,476 real windows; prevalences are 6.3602%, 0.83447% and 0.055173%. All probability metrics use calibrated predictions. High-threshold events are rare, so interpret accuracy together with PR-AUC and probability quality.

Threshold mm	Final model	Data	ROC-AUC	PR-AUC	Brier
0	XGBoost	Training	0.88029	0.61097	0.0341990
0	XGBoost	Validation	0.87681	0.58363	0.0347079
0	XGBoost	Test	0.87908	0.60809	0.0364909
5	XGBoost	Training	0.88578	0.23066	0.0069161
5	XGBoost	Validation	0.89447	0.23392	0.0070515
5	XGBoost	Test	0.87952	0.21839	0.0072009
25	XGBoost	Training	0.93661	0.03887	0.0005609
25	XGBoost	Validation	0.90912	0.03744	0.0007191
25	XGBoost	Test	0.90435	0.02089	0.0005449

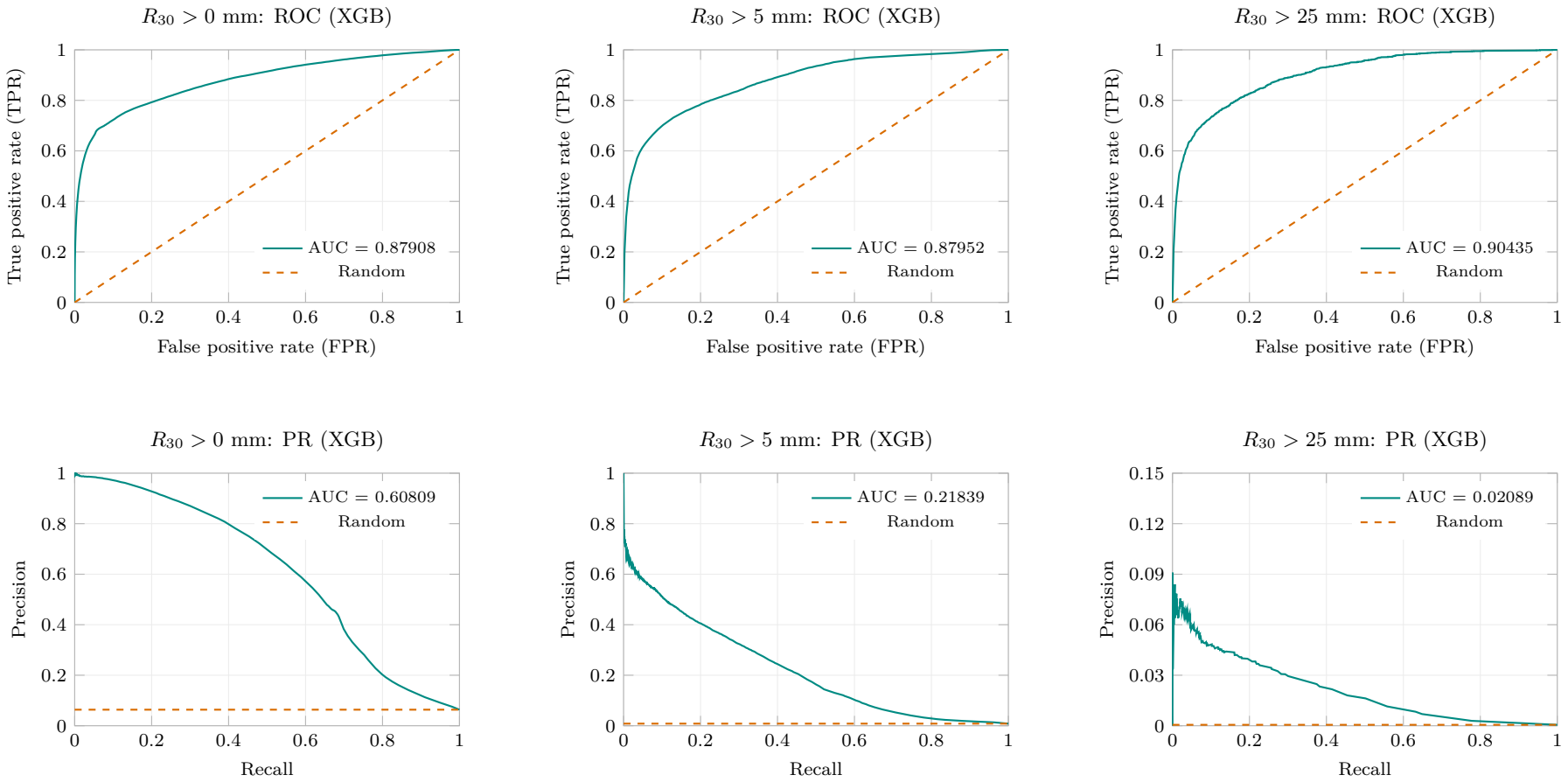
Training and test comparison. Training diagnostics use all real training observations, excluding SMOTE rows. Validation and test also use all eligible records. Cross-period PR-AUC differences reflect both generalization and changes in event prevalence.

Figure 8: test PR-AUC comparison



Test comparison. The selected models achieve test PR-AUCs of 0.60809, 0.21839, 0.02089. All exceed their random-ranking references (orange dashed lines). The low PR-AUC at 25 mm leaves substantial scope to improve heavy-rain precision and recall.

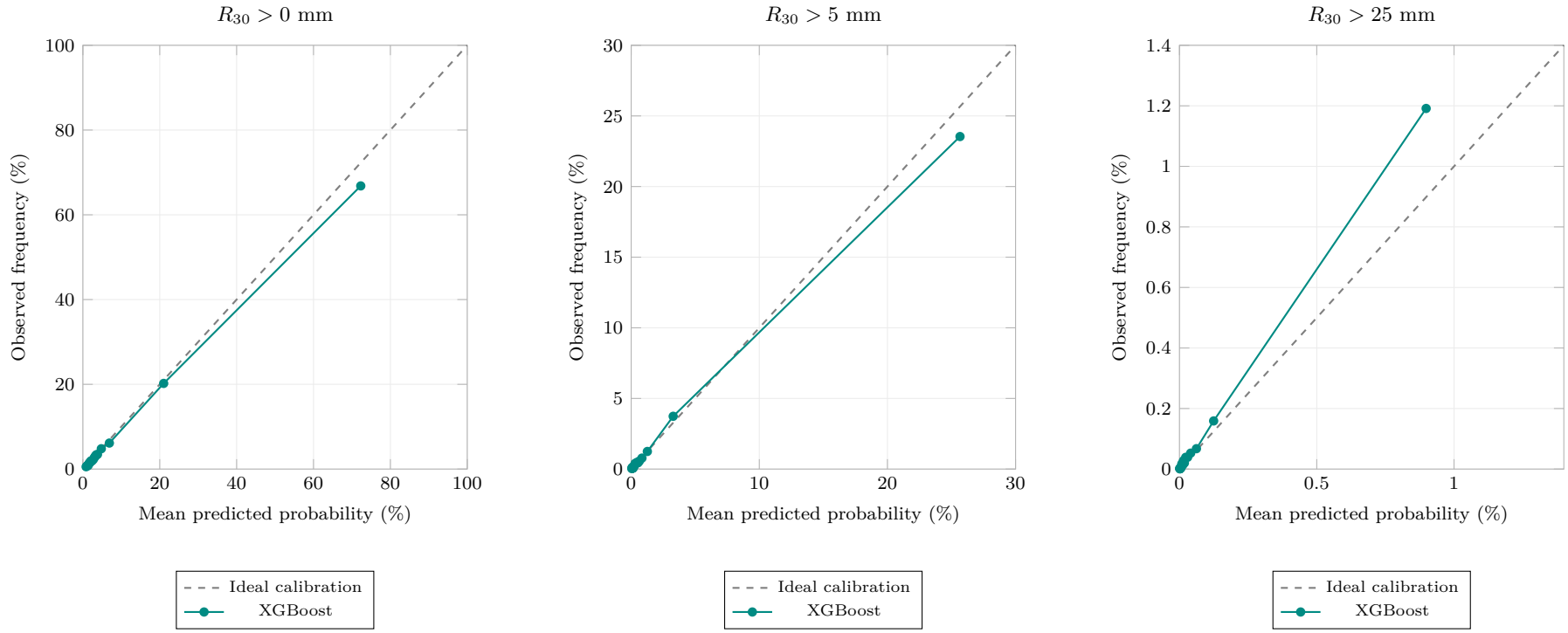
Figure 9: ROC and PR curves of the final selected models



Evaluation scope. Columns correspond to the 0, 5 and 25 mm rainfall thresholds, with a separately fitted classifier per target: 0 mm: natural-distribution XGBoost; 5 mm: class-weighted XGBoost; 25 mm: class-weighted XGBoost. Models are selected using 2023 H2 validation. All models fit the complete 34,542,285 eligible training records from 2017–2022. Curves and AUCs use all 6,374,476 eligible 2024 test windows. Dashed references are $y = x$ for ROC and test prevalence for PR. The 25 mm PR panel magnifies precision to 0–0.15.

Ranking performance. ROC-AUCs are 0.87908, 0.87952, 0.90435, and PR-AUCs are 0.60809, 0.21839, 0.02089. Higher rainfall thresholds describe rarer events. PR curves assess whether models achieve high precision and recall together.

Figure 10: reliability and probability quality



Probability quality. Mean forecast probabilities are 6.9640%, 0.8596%, 0.0422%, versus observed frequencies of 6.3602%, 0.8345%, 0.0552%. Overall, the 0 and 5 mm targets slightly overestimate risk, while the 25 mm target underestimates it. Each point aggregates similar forecasts; the dashed $y = x$ line represents agreement between forecast and observed frequency.

Calibration effect. Calibration reduces Brier scores for the class-weighted XGBoost models at 5 and 25 mm, correcting probability shifts induced by weighting. For 0 mm, Brier changes from 0.0363148 to 0.0364909; calibration does not improve test probability error for this target.

Rare-event calibration. For 25 mm, the highest-mean-probability group contains 164,015 windows: mean forecast 0.898% versus observed frequency 1.191%. The observed frequency exceeds the forecast, indicating underestimated heavy-rain probability in the higher-risk windows.

9. Classification metrics and further analysis

Threshold mm	Model	Cutoff	Precision	Recall	F1	Accuracy
0	XGBoost	0.3236	0.62066	0.56419	0.59108	0.95035
5	XGBoost	0.1758	0.29634	0.33540	0.31466	0.98781
25	XGBoost	0.0240	0.04208	0.16548	0.06709	0.99746

Classification evaluation. Cutoffs are validation-selected. Negative cases dominate high-threshold targets, so high accuracy alone does not establish effective heavy-rain detection. Compare ROC-AUC and PR-AUC primarily, with Brier and reliability for probabilities.

Threshold mm	Real positives	Added rows	Augmented positives (%)	Changed rain rows	Isolated copies
0	2,074,430	6,042,534	20.000	3,615,180	263,540
5	279,801	2,518,209	7.550	1,292,705	291,565
25	19,787	178,083	0.570	67,189	54,652

Spatial and temporal dependence. Neighbouring stations can share a rainfall event and half-hour windows overlap. Event/day-grouped evaluation can assess stability across stations and events. Independent threshold classifiers also motivate future monotone cross-threshold probability constraints.

10. Additional experiment: 5 mm tuning and prediction interface

Design. Candidates used all 27,434,009 eligible 2017–2021 records, with 2022 for tuning. Preprocessing was fitted only on the corresponding training block; future targets crossing split boundaries were purged. Six fixed configurations were ranked by PR-AUC, with Brier as a tie-breaker, without subsampling or early stopping. The selected configuration was refitted on all 34,542,285 eligible 2017–2022 records. Monotone Platt calibration used 2023 H1; probability-pipeline and cutoff selection used 2023 H2. The model was then frozen for 2024 evaluation. **Parameters.** Balanced XGBoost used 200 trees, depth 6, learning rate 0.05, 128 bins, minimum child weight 20, L2 regularisation 5 and L1 regularisation 0. The 2022 PR-AUC increased from 0.24617 for the original parameter configuration to 0.25049. The new procedure assigns missing indicators to every numeric feature, rather than selecting indicators as in the original procedure; gains cannot be attributed entirely to parameter tuning. Tables and curves in Sections 8–9 remain results of the original comparison experiment.

2024 test metric	Original comparison: 5 mm	Tuned final 5 mm model
PR-AUC	0.21839	0.22640
ROC-AUC	0.87952	0.87908
Brier score	0.0072009	0.0071609
Precision	0.29634	0.29402
Recall	0.33540	0.34657
F1	0.31466	0.31814

Interpretation. The test population contains 6,374,476 eligible windows, with 53,193 positives (0.83447%). PR-AUC increased by approximately 3.7% relatively and Brier decreased slightly, while ROC-AUC and precision declined slightly. Classification metrics use each procedure's validation-selected cutoff, not a common false-alarm rate. The new cutoff is 0.169824; mean predicted probability is 0.87725%. Brier is below the constant training-prevalence baseline of 0.0082751. Event-block confidence intervals were not computed, so statistical significance is not established. **Environment.** Features constructed on the local Mac were uploaded to a Google Compute Engine e2-highmem-8 VM: 8 vCPUs, 64 GB configured RAM (approximately 62 GiB reported), 100 GB Balanced Persistent Disk and Ubuntu 22.04.5 LTS x86_64. Software used Python 3.11, Java 17, Spark 3.5.8 and XGBoost 3.2.0. Spark ran in local[8] mode, with two XGBoost workers and four CPUs per task, a 6 GiB driver heap and no GPU. This describes the additional tuning, refitting, calibration and test experiment, not the original Task 1 or model-comparison runs.

10.1. Rejected spatial interactions and application limits

Spatial ablation. With the selected parameters fixed, eight interactions were derived from existing historical inputs: neighbour/own latest and half-hour rain gaps, neighbour peak/mean gap, neighbour wet fraction while the own station is dry, reading-age difference, own recent-15-minute rain share and log neighbour rainfall. Missing inputs remain missing; no future observations are used. These features do not measure wind direction or storm velocity.

Validation metric	Tuned model	Additional spatial interactions
2022 PR-AUC	0.250486	0.250632
2023 H2 PR-AUC	0.239991	0.238817
2023 H2 Brier	0.00701053	0.00701540

Selection. Ranking and probability error did not improve on the more recent validation block, so the spatial extension was rejected without using 2024 results to select it. The tuned model remains the 5 mm predictor. The original experiment had already inspected 2024; although the new parameter selection did not read it, this is not an entirely unseen holdout for the researchers. **Prediction interface.** Station, five-minute-grid prediction time and a 0, 5 or 25 mm threshold produce an exceedance probability. The tuned calibrated pipeline is the default for 5 mm; the original validation-selected models remain for 0 and 25 mm. An explicit option selects original models for comparison. Inference still requires a prepared archived feature row and is not a real-time service; historical row eligibility depends on complete future labels. Independent targets have no guaranteed cross-threshold probability ordering. Handling unknown station codes does not establish accuracy at new stations. **Additional AI use.** Codex assisted supplementary experiment scripts, cloud deployment, artifact checks and bilingual report editing. Metrics come from actual full-data runs; the spatial comparison comes from cloud validation output. Group members are responsible for review, and artificial fixtures are used only for implementation checks.

11. AI declaration and references

AI-use declaration. We use OpenAI Codex to assist study design, data-processing and model code, result checks and report editing. Reported rainfall statistics and model evaluations come from actual processing and model runs; SMOTE examples are used only for training.

Sources.

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2. Meteorological Service Singapore, Climate of Singapore.
3. NEA, Review of the climate in 2019. External context on dry conditions and the positive Indian Ocean Dipole, not an input feature in the proposed model.
4. geoBoundaries, Singapore ADM0, based on URA 2016 data, ODbL 1.0.
5. Apache Spark 3.5.8, MLlib classification and regression documentation, for logistic and linear SVM implementation. Evaluation uses Spark BinaryClassificationEvaluator.
6. XGBoost, PySpark estimator documentation, for external Spark integration; the actual installed version is 3.2.0.
7. scikit-learn, Precision–Recall documentation and probability-calibration documentation.